Safety Immersion and Gamified Hazard Training (SIGHT)

Research Assistant Packet

AR session (Session 2): Magic Leap 2 headset at the real machines · standing

Participant ID	**P** _____	**Session date**	_____
Machine order	Order ____ (as in Session 1; see Overview #1)		
Research assistant	_____	**Start time**	_____

Every script and research assistant form for one AR session, in the order you use them. The participant's form (the TAM questionnaire) is a separate PDF. Print both single-sided and do not staple. There is no consent form in Session 2: it was signed in Session 1.

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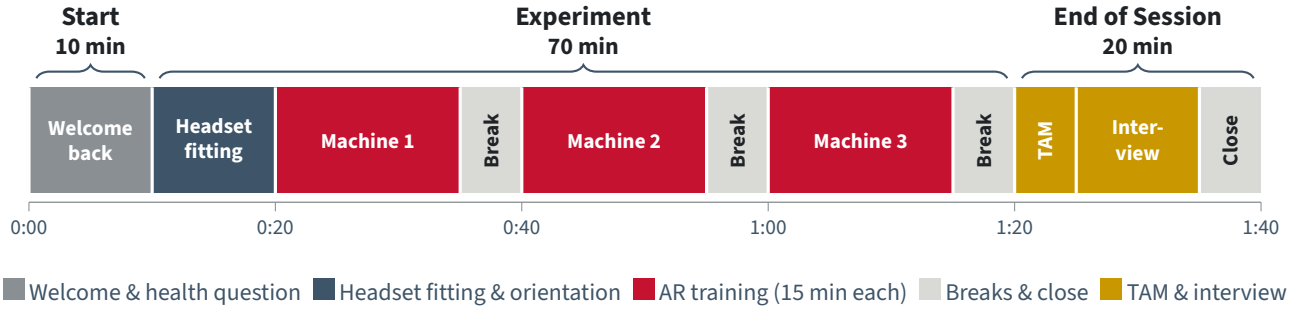
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P **Participant forms** (separate PDF): End of Session #1.

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Session at a Glance

Schematic of the AR session, with planned minutes for every step



Planned total **about 100 minutes**. No consent form, baseline questionnaires or machine assessment in Session 2. The times are estimates until the pre-pilot.

This participant

Participant ID	**P** _____ (order _____)
Machines	_____ → _____ → _____
Order	the same as in Session 1
Headset	Magic Leap 2

Materials

- ☐ This packet and the participant forms, printed single-sided, unstapled
- ☐ Magic Leap 2 headset, charged; antibacterial wipes
- ☐ Printed tags in place at each machine
- ☐ Pens, clipboard, timer (15:00 per machine)
- ☐ Audio recorder, set to m4a or mp3 (only if audio consent applies)
- ☐ $75 Visa gift card for the participant

Contacts

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Machine orders (the same as in Session 1)

Order	1st	2nd	3rd
1	Mill	CNC Lathe	Cobot
2	Mill	Cobot	CNC Lathe
3	CNC Lathe	Mill	Cobot
4	CNC Lathe	Cobot	Mill
5	Cobot	Mill	CNC Lathe
6	Cobot	CNC Lathe	Mill

Participant n uses order $((n-1) \bmod 6) + 1$: P1 and P7 use order 1, P2 and P8 order 2, and so on. A = Mill, B = CNC Lathe, C = Cobot.

Key rules

- Do the machine checks (Experiment #1) before the participant arrives. A machine that fails a check is not used.
- A researcher stays near the participant at all times.
- Stop each machine at 15:00. Give a 5-minute break after each machine.
- Dizziness, nausea, eye strain or headache: stop, headset off, sit down, stay in the lab until it passes.
- Never write a participant's name on any page.

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Research Assistant Run Sheet

Keep this sheet beside you. Tick each step and write the clock time.

Before the participant arrives

- ☐ Check that every page of this packet and of the participant forms is present (see each cover).
- ☐ Do the machine checks on Experiment #1 and sign them. A machine that fails a check is not used.
- ☐ Headset charged and wiped; printed tags in place at each machine; pens and timer ready.

	Planned	Clock time	Step	Document
Start of Session				
☐	0:00		Welcome back, rights, same participant ID (script on the next page)	This sheet
☐	0:05		How they felt after Session 1; health question asked again	This sheet
Experiment				
☐	0:10		Headset fitting and orientation (about 10 min)	Experiment #1
☐	0:20		Machine 1: _____ (15 min, timed)	Experiment #2
☐	0:35		5-minute break	
☐	0:40		Machine 2: _____ (15 min, timed)	Experiment #3
☐	0:55		5-minute break	
☐	1:00		Machine 3: _____ (15 min, timed)	Experiment #4
☐	1:15		5-minute break	
End of Session				
☐	1:20		TAM questionnaire, rating the AR system (about 5 min)	End of Session #1 P
☐	1:25		Interview (about 10 min)	End of Session #2
☐	1:35		Close out ($75 Visa gift card; headset back and wiped); data entry within 24 hours	End of Session #3

P = in the participant forms PDF; every other document is in this packet.

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Start of Session scripts

Welcome back and rights. Thank the participant for coming back. Remind them of their rights, as in Session 1.

> **SAY** *Welcome back, and thank you for coming to the second session. As last time, taking part is voluntary: you may skip any question, and you may stop at any time without penalty. Today you will use the SIGHT AR system, with a headset, at the real machines.*

Same participant ID. Use the same ID as in Session 1: **P** _____. The consent form signed in Session 1 covers this session; there is no new consent form. Never write the participant's name on any page.

How did they feel after Session 1?

> **SAY** *How did you feel after the first session? Did you have any dizziness, headache, nausea, or eye strain afterward?*

Notes: _____

Health question, asked again (protocol 1.1). Read it word for word.

> **SAY** *Do you have any condition, such as vision abnormalities, pregnancy, a heart condition, or illness that includes headache, nausea, or dizziness?*

Answer today: ☐ NO ☐ YES

If the answer is YES, the person cannot take part today: thank them and end the session. Give the $75 Visa gift card, as for any session that ends early.

No questionnaires at the start of Session 2. The demographic form, the JHA quiz and the GATT were completed in Session 1. Go to Experiment #1.

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Machine Checks & Orientation

Machine checks before the participant arrives; then about 10 minutes of headset fitting and orientation

Machine checks (before the participant arrives)

- ☐ Mill: the circuit breaker is off, and the spindle is confirmed not to start.
- ☐ CNC lathe: the door (guard) is closed. The lathe runs only with it closed.
- ☐ Cobot: reduced speed is set, collision detection is on, and the AR shows the no-go zone.
- ☐ A trained team member is at the emergency stops: _____

Checked by (initials) _____ Time _____

If a check fails

A machine that fails a check is not used in this session. Skip it, write why on its observation sheet, and tell the PI.

Headset fitting and orientation

- ☐ Fit the Magic Leap 2 headset; the participant can see clearly.
- ☐ Show the hand gestures for each step.
- ☐ Show how the printed tags anchor each step to the machine.
- ☐ Show how to realign the overlay if it drifts.
- ☐ Check for dizziness or eye strain before starting: none.

SAY *This is the SIGHT AR safety training system. You will train on each of 3 machines: lathe, mill, and collaborative robot. The headset will guide you step by step at each machine. If anything is confusing or difficult, try to work through it as you normally would. Please stay out of the zone the headset shows near the robot, and tell me right away if you feel dizzy, sick, or have eye strain or a headache.*

Stop rule

A researcher stays near the participant at all times. A participant who reports dizziness, nausea, eye strain or headache stops, takes off the headset, sits down, and stays in the lab until the effects pass. Note it on the observation sheet.

How much help to give (all machines)

1. Observe silently.
2. Safety or technical support only (e.g., the headset loses the overlay).
3. One neutral prompt if stuck: *"Please continue in the way that feels most natural to you."*
4. If asked for help: *"I am not able to guide you, but you may use any instructions in the headset."*

Log each request on the observation sheet.

RESEARCH ASSISTANT ONLY · Do not hand this page to the participant Participant ID: **P** _____

Machine 1 of 3: _____

Timed AR training: 15 minutes. The participant does not have to finish every module.

Start time _____ 15:00 mark _____ Observer _____

☐ Stopped at 15:00 ☐ Finished early at _____ ☐ Participant stopped early (note why)

> **SAY** *Please follow the guidance in the headset for the _____. You will have about 15 minutes to work through as much of the training as you can. Do not worry if you do not finish every step.*

Safety at this machine: Mill: breaker off, the spindle cannot run. Lathe: runs only with its door closed. Cobot: reduced speed; the participant stays out of the zone the headset shows.

Modules reached

AR training module	Completed	Clock time and notes
Module 1	☐	
Module 2	☐	
Module 3	☐	
Module 4	☐	
Module 5	☐	
Module 6	☐	
Module 7	☐	
Module 8	☐	

Help requests and points where the participant got stuck

Time	What happened (type of difficulty)	Help level given
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none

Discomfort: ☐ None ☐ Fatigue ☐ Pain ☐ Eye strain ☐ Dizziness ☐ Loss of balance ☐ Other **Any: stop**

Next: 5-minute break (restroom, water, walk around), then Experiment #3, Machine 2.

Observation notes: confusion, frustration, and the participant's comments word for word

RESEARCH ASSISTANT ONLY · Do not hand this page to the participant Participant ID: **P** _____

Machine 2 of 3: _____

Timed AR training: 15 minutes. The participant does not have to finish every module.

Start time _____ 15:00 mark _____ Observer _____

☐ Stopped at 15:00 ☐ Finished early at _____ ☐ Participant stopped early (note why)

> **SAY** *Please follow the guidance in the headset for the _____. You will have about 15 minutes to work through as much of the training as you can. Do not worry if you do not finish every step.*

Safety at this machine: Mill: breaker off, the spindle cannot run. Lathe: runs only with its door closed. Cobot: reduced speed; the participant stays out of the zone the headset shows.

Modules reached

AR training module	Completed	Clock time and notes
Module 1	☐	
Module 2	☐	
Module 3	☐	
Module 4	☐	
Module 5	☐	
Module 6	☐	
Module 7	☐	
Module 8	☐	

Help requests and points where the participant got stuck

Time	What happened (type of difficulty)	Help level given
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none

Discomfort: ☐ None ☐ Fatigue ☐ Pain ☐ Eye strain ☐ Dizziness ☐ Loss of balance ☐ Other **Any: stop**

Next: 5-minute break (restroom, water, walk around), then Experiment #4, Machine 3.

Observation notes: confusion, frustration, and the participant's comments word for word

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Machine 3 of 3: _____

Timed AR training: 15 minutes. The participant does not have to finish every module.

Start time _____ 15:00 mark _____ Observer _____

☐ Stopped at 15:00 ☐ Finished early at _____ ☐ Participant stopped early (note why)

> **SAY** *Please follow the guidance in the headset for the _____. You will have about 15 minutes to work through as much of the training as you can. Do not worry if you do not finish every step.*

Safety at this machine: Mill: breaker off, the spindle cannot run. Lathe: runs only with its door closed. Cobot: reduced speed; the participant stays out of the zone the headset shows.

Modules reached

AR training module	Completed	Clock time and notes
Module 1	☐	
Module 2	☐	
Module 3	☐	
Module 4	☐	
Module 5	☐	
Module 6	☐	
Module 7	☐	
Module 8	☐	

Help requests and points where the participant got stuck

Time	What happened (type of difficulty)	Help level given
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none
		1 2 3 4 none

Discomfort: ☐ None ☐ Fatigue ☐ Pain ☐ Eye strain ☐ Dizziness ☐ Loss of balance ☐ Other **Any: stop**

Next: 5-minute break (restroom, water, walk around), then End of Session #1, the TAM questionnaire (in the participant forms).

Observation notes: confusion, frustration, and the participant's comments word for word

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Post-Session Interview

Read each question aloud, then write the participant's answer and key quotes

First, the TAM (End of Session #1, in the participant forms). Hand it over; it rates the SIGHT AR system.

> **SAY** *Now, please complete this survey. It will ask about your experience using the system. You may leave questions blank that you do not want to answer. Then, let's talk a little bit about what made the system easy or difficult to use. If you agree, I will audio record our interview.*

Audio recording

☐ Participant agreed: recording started at _____ ☐ Participant declined: notes only

Record only if the participant agreed on the consent form in Session 1 (see this ID's VR interview page) and agrees again today. Otherwise, write the answers on these pages. When recording on the study recorder, still note key points and the recorder time stamp beside them.

The audio choice is on the revised consent form, which is still a draft. Do not record until the IRB approves it.

1. If you have taken any other machine safety training courses, how does the SIGHT AR system compare to those training courses?

2. What did you enjoy most about the training modules that you worked through?

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3. Do you have any suggestions on how to improve the SIGHT system to make it more effective and/or enjoyable to use? This can be related to the interface, the specific training content, or the way that the training content is presented.

4. Are there aspects of the SIGHT system that we should make sure to keep in any future revisions?

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Close-out & Data Entry

Close the session, then upload everything within 24 hours

Close the session

- [ ] Thank the participant for taking part in both sessions.
- [ ] Check how the participant feels. Anyone with lingering effects from the headset stays at the lab until they pass.
- [ ] Take the headset back and wipe it with antibacterial wipes before the next participant.
- [ ] Give the participant a **$75 Visa gift card** for this session. If they sign a receipt, keep it with the signed consent forms (it includes their name), not in this packet.

Step 1. Archive in Google Drive (shared **SIGHT Pilot Testing** folder): the packet and transcript (never the audio) go in this participant's **AR** subfolder.

```
SIGHT Pilot Testing/
├─ Signed Consent Forms/  ← nothing to add in Session 2
└─ P _____/  ← one folder per participant
   ├─ AR/  ← this session
   └─ VR/
```

- [ ] Scan the participant forms (the TAM) and this packet together as one PDF and upload it to the **AR** subfolder:
  `P _____AR_packet_YYYY-MM-DD.pdf`
- [ ] Upload the transcript, if recorded: `P _____AR_interview_YYYY-MM-DD_transcript.txt`

Step 2. Enter the data in the SIGHT Test Hub (`sight-test-hub.vercel.app`; sign in with your Miami Google account)

- [ ] Upload the same packet PDF. If the Hub cannot identify a page, it asks you first.
- [ ] If recorded: transcribe the interview on the SIGHT workstation with the **SIGHT Interview Transcriber**, check it against the recording, then delete the recording from the recorder and the workstation.
- [ ] Upload the transcript file (names as [name]). In **Interview Transcript**, confirm the four answers and tick that the recording is deleted. Then delete the transcript files from the workstation.
- [ ] Open each page, compare every field with the paper, correct anything misread, and confirm the page. Black out any name written on a page.
- [ ] Ask a team member who confirmed none of the pages or the transcript to check the session. Upload the transcript first: one added after the check sends the session back to review.
- [ ] No hand scoring: the Hub scores the TAM.